

Yuzhuo Jia

Ph.D. Student · Kyushu University

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Research Interests

My research lies at the intersection of agentic AI, large language models, and human-computer interaction. I focus on agent-to-agent interaction, embodied cognition, and AI-driven learning systems.

Education

Oct 2026 – Oct 2029	Kyushu University Ph.D. Student, Department of Information Science and Technology Fukuoka, Japan · Expected graduation: October 2029
Feb 2023 – Feb 2025	University of Sydney M.Sc. Computer Science Sydney, Australia

Publications

2026	Human and LLMs' Bodily Representations Jia, Y. [*] , Yu, K. [*] , Zhan, X., Zhang, Y.-Q., & Zheng, C. <i>Nature Machine Intelligence</i> · Submitted
2026	Balancing Automation and Agency: Designing Adaptive Reading Interfaces for Middle-Aged Office Workers Yu, K. [*] , Jia, Y. [*] , Fan, H., Zheng, C., & Peng, Z. <i>ACM CHI Conference on Human Factors in Computing Systems (CHI)</i> · Poster · Accepted
2026	LUMOS: Designing an Interactive Companion Robot to Enhance Critical Reading and Thinking in Academic Contexts Mao, Y. [*] , Li, Y. [*] , Jia, Y., Li, F., Yin, Z., & Zheng, S. <i>35th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)</i> · Accepted
2024	FODAP Graph for Enhanced Medical Imaging Narrative Generation Shu, K. [*] , Jia, Y. [*] , Zhang, Z., & Gao, J. <i>International Conference on Bioinformatics and Biomedicine (BIBM)</i> · Accepted [DOI]
2024	Pedestrian Behavior Detection and Traffic Violation Recognition Based on YOLOv5 Jia, Y. <i>4th International Conference on Image Processing and Intelligent Control</i> · Accepted [DOI]

^{*} Equal contribution.

Research Experience

Tsinghua University

Research Assistant · Beijing, China

Sep 2026 –
Present

MatMem: Materialized Latent Memory for LLM Agents

- Developed a latent-memory framework that stores an agent's prior computation and tool state as compact key-value (KV) slots. Trained a memory writer with learnable memory tokens and position-specific LoRA, while keeping the base model frozen during reading.
- Built controlled tasks and evaluation pipelines for computed-state recall, interrupted-task resumption, and long-context memory. Compared text notes, KV compression, and latent-memory baselines across Qwen and Llama models, including Stateful GSM-Symbolic and τ -bench evaluations.
- Achieved **94.0%** accuracy on an eight-account ledger task with 16 memory slots (five-seed mean). Memory writing was **8–17 $\times$ faster** than summary generation in controlled A100 experiments.

Sep 2025 –
Sep 2026

Human and LLMs' Bodily Representations

- Investigating whether pretrained LLMs exhibit human-like bodily representations in virtual environments, spanning conceptual knowledge and sensorimotor experience. Helped design human-LLM comparison tasks, informed by a literature review in embodied cognition.
- Built an analysis framework for open-source vision-language models: model deployment, layer-wise embedding extraction, PCA and t-SNE visualization, and representational similarity analysis (RSA) to compare human and model representations.
- Contributing experimental design, computational modeling, data analysis, and the Methods and Results sections to the manuscript.

Jun 2025 –
Sep 2025

Adaptive Reading System

- Designed and implemented a Tampermonkey reading interface that uses MediaPipe Face Mesh to estimate viewing distance and adapt font size and color contrast in real time.
- Conducted a study with **24 middle-aged adults** across four reading conditions, measuring speed, comprehension, and comfort. Thematic analysis of semi-structured interviews identified trade-offs between adaptation-related cognitive load and performance gains.
- Co-authored the Related Work, System Design, and Discussion sections of the paper.

Apr 2025 –
Aug 2025

Embodied Intelligence Research Group

- Built a 6-DOF force-controlled robotic arm using 3D-printed components and DAMIAO motors; integrating MoveIt 2 for motion planning and control.
- Extended an open-source web agent framework with speech-to-text, real-time video streaming, and a Model Context Protocol (MCP) client for compatible tools and services.

Jan 2025 –
Mar 2025

PI Lab – MathAdventure

- Contributed to a screen-free voice learning system for children aged 4–8, grounded in Contextual Learning and the Zone of Proximal Development (ZPD). Built real-time recognition of everyday objects, such as cups and spoons, to generate contextual decimal-system problems.
- Conducted Wizard of Oz studies, iterative prototype tests, and structured interviews with **18 child-parent pairs**, evaluating engagement, learning effectiveness, and ZPD-based scaffolding.
- Contributed Related Work and System Design and Development to “MathAdventure: Contextual and Zone of Proximal Development-driven Math Learning through Symbolizing Real-World Objects with the Decimal System.”

University of Sydney

Sydney, Australia

Jun 2024 –
Dec 2024

BiFocalNet: Dual-Branch Architecture for Enhanced Remote Sensing Segmentation

- Proposed BiFocalNet for remote sensing segmentation, achieving state-of-the-art performance on GID (+**3.06 mIoU** over prior work).
- Designed an EfficientNetV2 + PVT dual-branch encoder for local details and global context, with Cross-Fusion and SuperASPP for multi-scale feature fusion.
- Ran extensive experiments and ablations; improved over ParaTransCNN by +3.28 (Forest) and +3.51 (Built-up) mIoU.

Mar 2023 –
May 2024

FODAP Graph for Enhanced Medical Imaging Narrative Generation

- Built a MedSAM-based ViT encoder for representations of 512×512 medical images.
- Removed the MLP neck and used patch embeddings directly, reducing features from 1024×768 to 256×768 to unify outputs and improve generalization.
- Compared BioMedCLIP-pretrained ViT-B/16@224 and ImageNet21k-pretrained CvT@384; MedSAM performed best.

Technical & Research Skills

Languages	Python, JavaScript, TypeScript, SQL, LaTeX
ML / Vision	PyTorch, MediaPipe, ROS 2
Web / Apps	React, Next.js, Electron, WebXR, Tailwind CSS, Node.js
Tools	Git, Unity, Arduino, SQL Databases, Figma
Methods	UI/UX Prototyping, Qualitative & Quantitative Analysis